Climate Resilience: A Multi-Source Analysis of Climate Policy Evolution Post Paris Agreement (2013–2025)

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This application report documents a data mining pipeline which brings together four separate sources of climate data: the OECD CAPMF, the Climate Policy Database (CPDB), the UNFCCC NDCs and the Notre Dame Global Adaptation Initiative (ND-GAIN).The exploration analyzes the evolution of climate policy strategies since the signing of the Paris agreement on the 4th of November 2016, with a particular focus on the integration of adaptation measures alongside mitigation efforts. The focus is on a selection of 12 countries, with an effort to include members from both the Global South as well as the Global North. By combining API access, web scraping and natural language processing (NLP), the pipeline quantifies the shift from mitigation to adaptation and reveals a 'Commitment Gap' between Global North and Global South countries.

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1 Introduction WIP

Climate policy research traditionally focuses on mitigation, with an emphasis on the reduction of greenhouse gas emissions. However, the increasing frequency and severity of climate impacts has elevated the importance of adaptation measures in national policy portfolios. The Paris Agreement (2015) marked a pivotal moment in global climate governance, with its dual focus on mitigation and adaptation, but measuring adaptation is inherently challenging. As the Grantham Research Institute notes, "unlike climate mitigation, where responses like carbon pricing and emission targets are widely tracked, adaptation laws and policies have received less systematic scrutiny" [1, p.10].

This measurement gap has real consequences. International climate databases systematically under represent adaptation policies, particularly in the Global South where adaptation is a **survival imperative** rather than a discretionary policy choice. When adaptation is invisible in the data, it becomes invisible in policy debates, climate finance allocation and accountability mechanisms. The lack of "comparative global analysis and systematic scrutiny of national adaptation legislation and policies has hampered the ability to assess progress, identify gaps, and understand effective frameworks" [1, p.10]. This project aims to bridge the measurement gap and serves as a digital audit of climate ambition across our selection of countries.

Comparing the Global North to the Global South is a major focus of this project. The Global North has historically been responsible for the majority of emissions and has more resources to invest in both mitigation and adaptation. The Global South, whilst being less responsible for the acceleration of climate change, are the ones who bear the brunt of it.

2 Research Question WIP

By combining multiple data sources and methodologies, we seek to answer the question: *Have climate policy portfolios shifted from mitigation toward adaptation after the Paris Agreement, and to what extent does this shift differ between Global North and Global South countries?*

Key Terms: - **Climate Policy Portfolio**: The collection of all climate-related policies enacted by a country. - **Mitigation**: Efforts to reduce or prevent the emission of greenhouse gases. - **Adaptation**: Efforts to adjust to actual or expected climate change and its effects - **Global North**: Generally refers to developed countries with higher income levels, often located in the Northern Hemisphere. - **Global South**: Generally refers to developing countries with lower income levels, often located in the Southern Hemisphere. - **Paris Agreement**: An international treaty adopted in 2015 that aims to limit global warming to well below 2 degrees Celsius, with efforts to limit it to 1.5 degrees Celsius. - **Climate Ambition**: The level of commitment and action a country demonstrates in its climate policies, often measured through the strength and scope of its mitigation and adaptation efforts. - **Commitment Gap**: The disparity in climate policy ambition and implementation. - **Responsibility Gap**: The disconnect between those who cause climate change and those who suffer its worst effects.

The research question is answerable with the described data mining approach because of the following: 1. **Temporal dimension**: Policy databases (CPDB) provide time-stamped records spanning 2013–2025, allowing pre/post-Paris comparison. 2. **Textual dimension**: NDC texts (publicly available PDFs) can be analyzed using NLP to quantify adaptation vs. mitigation emphasis. 3. **Geographic dimension**: Countries can be stratified by climate vulnerability (ND-GAIN Index) and development status (Global North/South).

Technical Decisions:

The Temporal Boundary (2013–2025) The choice of 2013 as the starting point is strategic. It allows us to capture the policy landscape leading up to the Paris Agreement, providing a baseline for comparison. The end point of 2025 ensures we have the most recent data available.

Country Selection The selection of 12 countries is designed to provide a diverse sample that includes both Global North and Global South nations. The logic is as follows:

Country	ISO-3 Code	Region	Climate Vulnerability (ND-GAIN)
Canada	CAN	North	Low
Germany	DEU	North	Low
Japan	JPN	North	Low

Country	ISO-3 Code	Region	Climate Vulnerability (ND-GAIN)
India	IND	South	High
South Africa	ZAF	South	High
Denmark	DNK	North	Low
United Kingdom	GBR	North	Low
Chile	CHL	South	Medium
Colombia	COL	South	Medium
United States	USA	North	Medium
Saudi Arabia	SAU	South	Medium
Rwanda	RWA	South	High

3 The Data Ecosystem WIP

This project brings together four distinct data sources, each with its own strengths and limitations:

3.1 Source 1: OECD CAPMF (Stringency Scores via API)

What it provides: Standardized "stringency" indicators that measure the 'hardness' of policy law (e.g., mandatory regulations score higher than voluntary guidelines). These indicators are available for a subset of OECD member countries and focus primarily on mitigation policies.

Access method: OECD R package with SDMX API queries.

Strength: High data quality; institutional credibility.

Limitation: Geographic bias toward OECD member countries (mostly Global North). Federal systems like the USA have reporting gaps because sub national policies aren't captured (documented in Issue #2: "Addressing Missing Data in OECD CAPMF"). The focus of this source is on mitigation policies so it doesn't capture the full picture of adaptation efforts, which are often less formalized and harder to track in standardized databases.

3.2 Source 2: NewClimate Institute's Climate Policy Database

What it provides: Policy-level data on the number and type of climate policies enacted by countries.

Access method: `cpdb-api` Python package, accessed from R via `reticulate` (R-Python bridge).

Strength: Includes labels for policy intent; Broader geographic coverage than OECD; includes many Global South countries.

Limitation: The data is not exhaustive. Policy coverage and information depth vary across jurisdictions, and sub national level policies are not comprehensively indexed. There is a strong bias toward Global North countries, with many Global South countries having sparse data (documented in Issue #1: "Pivot from Absolute Policy Counts to Adaptation Ratio")

3.3 Source 3: UNFCCC NDCs

What it provides: Official Nationally Determined Contribution (NDC) pdfs containing countries' climate commitments, including qualitative descriptions of adaptation priorities and legal language (mandatory vs. aspirational).

Access method: Automated pipeline: 1. Access the csv from Open Climate Data Github Repo which contains links to each pdf. 2. Filter for our target list and access only the most recently published documents. 3. Download to local cache. 4. Extract text using `pdftools` R package.

Strength: Primary source documents; unmediated access to government priorities. The NDCs are the official statements of climate ambition and are critical for understanding how countries frame their commitments. They provide rich qualitative data that can be analyzed for both content (adaptation vs. mitigation focus) and tone (the "hardness" of language used).

Limitation: NDCs are diplomatic pledges rather than tangible laws, so they may not always reflect actual policy implementation. The language can be vague or aspirational, making it challenging to quantify. Additionally, the NDCs are updated periodically, and there may be a lag between the latest data available and the current policy landscape. The European Union's NDC is a single document that represents multiple member states, which complicates country-level analysis.

3.4 Source 4: ND-GAIN Index (Vulnerability Benchmarking via Manual CSV)

What it provides: Composite vulnerability and readiness scores for each country, measuring exposure to climate hazards and institutional capacity to adapt.

Access method: Manual CSV download.

Strength: Globally comparable vulnerability metric; multidimensional assessment; major asset for classifying countries by climate risk and adaptation need.

Limitation: Manual download reduces reproducibility; temporal lag (most recent data from 2023)

3.5 Why Multiple Sources?

No single data source provides a complete picture of climate policy evolution. By triangulating across multiple sources with different strengths and weaknesses, we can build a more robust and nuanced understanding of how climate policy portfolios have evolved since the Paris Agreement. The OECD CAPMF provides standardized stringency scores for mitigation policies, while the CPDB offers a broader view of policy adoption across both mitigation and adaptation. The NDCs give us direct insight into how countries frame their commitments and the ND-GAIN Index allows us to contextualize these commitments against vulnerability and readiness.

4 The Technical Pipeline WIP

The entire workflow is orchestrated by a single master script (`main.R`) that manages: 1. **Dependency installation** 2. **Data acquisition** 3. **Data processing** (cleaning, merging, feature engineering). 4. **Analysis** 5. **Output generation** (figures, tables, this report). 6. **Environment Cleaning** (removing temporary files and data).

5 Results WIP

5.1 Finding 1.1: The Paris Inflection (Accelerated Policy Adoption)

I started off my analysis with a simple plot of climate policy growth to see if there has been a change in the amount of climate policies implemented by our target countries since the Paris Agreement. This is a basic starting point to understand the overall trajectory of climate policy evolution.

Visualization: Figure 1 shows cumulative policy adoption over time for our 12 countries, with a vertical line at December 2015 (Paris Agreement).

Observation: Most countries show a visible acceleration in policy adoption around the time of the Paris Agreement, particularly in the Global North.

Interpretation: The Paris Agreement appears to have catalyzed increased policy activity, consistent with the hypothesis that international agreements influence domestic legislation. However, this could also reflect improved data collection by CPDB (a limitation discussed below)

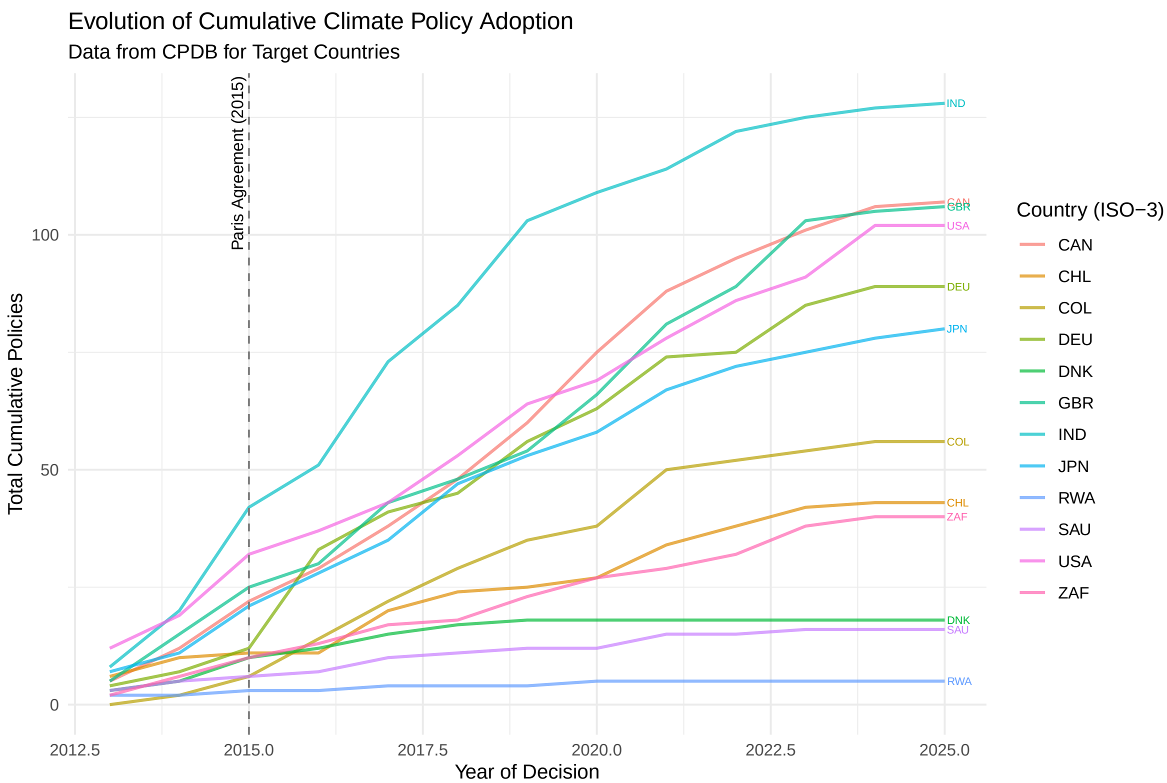

Figure 1: Cumulative Climate Policy Adoption by Target Country (2012-2026)

5.2 Finding 1.2: The Global Adaptation Growth

To understand the evolution of adaptation policies specifically, I filtered the CPDB data for policies labeled as “adaptation” or “both” (mitigation + adaptation) and plotted their cumulative growth over time. This allows us to see if there has been a similar inflection point for adaptation policies post-Paris. **Visualization:** Figure 2 shows the cumulative number of adaptation policies globally from 2013 to 2025. **Observation:** There is a clear upward trend in the number of adaptation policies, with a noticeable increase starting around 2015-2016. This suggests that the Paris Agreement may have also spurred growth in adaptation policy adoption.

Interpretation: The Paris Agreement’s recognition of adaptation as a key pillar of climate action likely contributed to this growth. However, the increase in adaptation policies is not as steep as the overall policy growth.

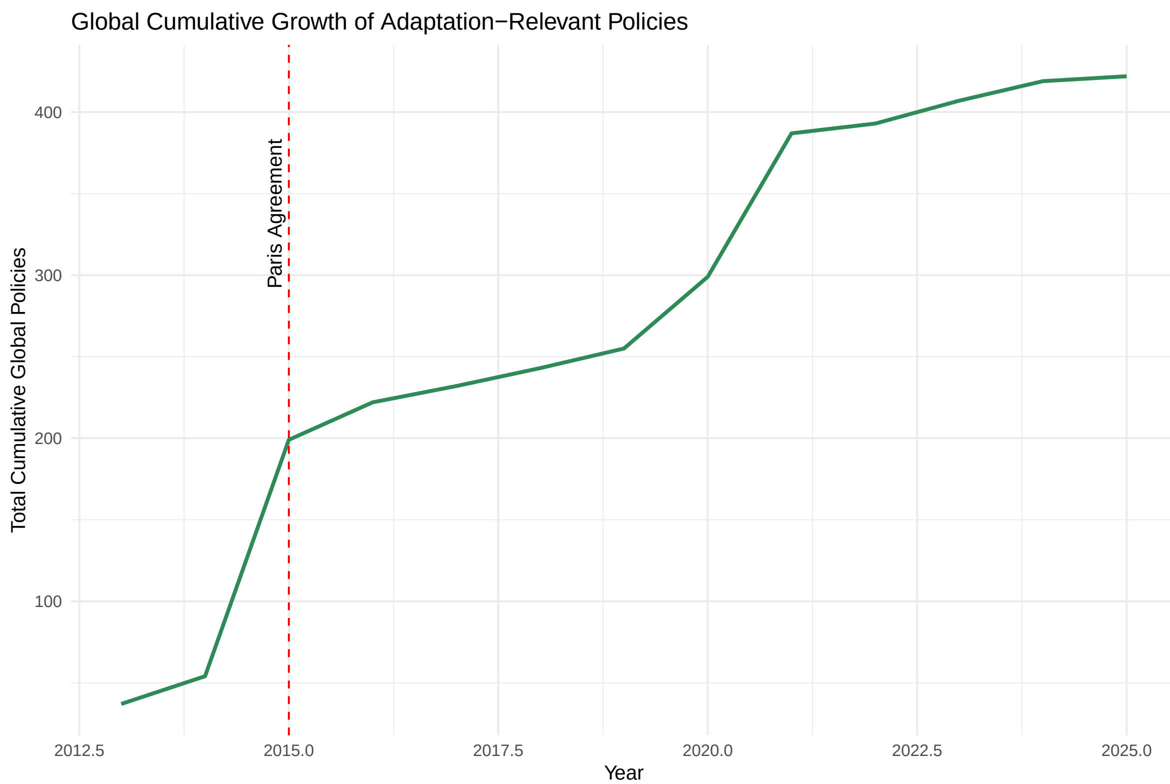

Figure 2: Global Cumulative Growth of Adaptation Policies (2013-2025)

5.3 Finding 2: The Survival Ratio (Vulnerability → Adaptation)

This finding served as the starting point for my deeper dive into the adaptation gap. I wanted to see if there was a correlation between a country's climate vulnerability (as measured by the ND-GAIN Index) and the proportion of their climate policy portfolio dedicated to adaptation measures (the "Adaptation Mix"). I did this by collecting data on the number of adaptation policies vs. total climate policies for each country from the CPDB, and then calculating the percentage of the portfolio that is focused on adaptation. I then plotted this against the ND-GAIN vulnerability scores to see if there was a relationship.

Visualization: Figure 3 shows Adaptation Mix (% of portfolio dedicated to adaptation) for each country, sorted by ND-GAIN vulnerability score.

Observation: Countries with high climate vulnerability (e.g., Rwanda, South Africa) have significantly higher Adaptation Mix compared to low-vulnerability countries (e.g., Germany, Denmark). This is consistent with the Grantham Research Institute's documentation that "all 26 development plans that incorporate adaptation at a significant level originate from 13 countries in the Global South" [1, p.14-15].

Interpretation: For highly vulnerable countries, adaptation is a survival imperative. As climate risk increases, legislative "mindshare" shifts from emissions reduction (mitigation) to resilience-building (adaptation).

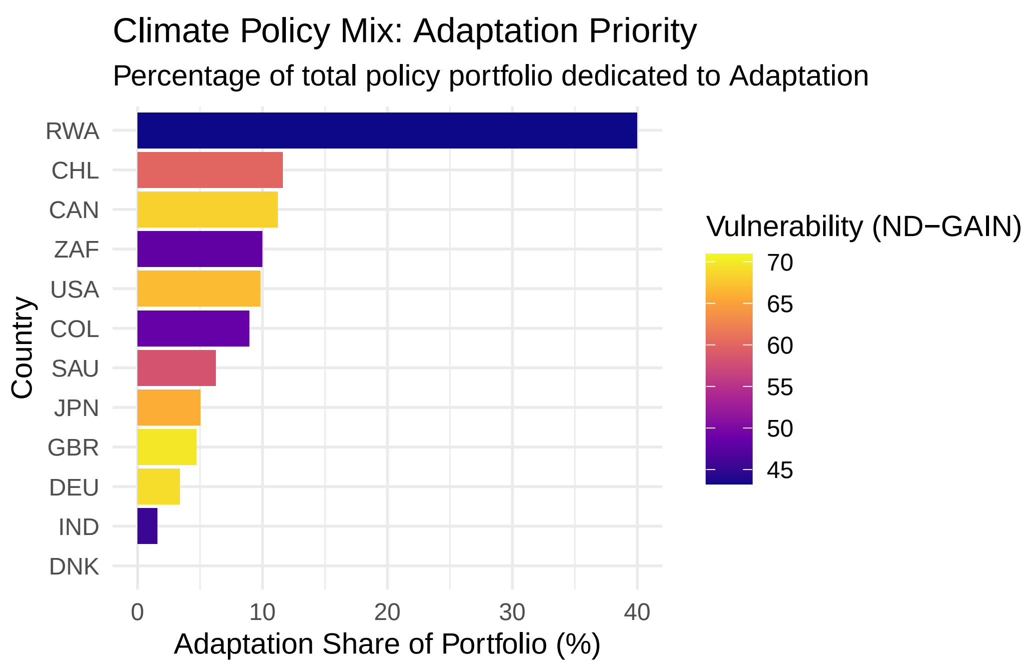

Figure 3: The Adaptation Mix: Percentage of Portfolio dedicated to Resilience

5.4 The Commitment Gap

The final finding is the most complex and nuanced. It emerged from my analysis of the NDC texts, where I used NLP techniques to quantify both the focus on adaptation vs. mitigation (keyword frequency) and the "hardness" of the language used (binding legal terms vs. aspirational language). I then plotted these two dimensions against each other to see if there were any patterns in how countries frame their climate commitments.

Visualization: Figure 4 (the **Ambition Matrix**) plots Lexical Hardness (y-axis) vs. Adaptation Focus (x-axis) for each country's NDC.

Observation: Global North countries cluster in the **low focus, high hardness** quadrant (they prioritize mitigation and use binding legal language). Global South countries cluster in the **high focus, low hardness** quadrant (they prioritize adaptation but use aspirational language).

Interpretation: This is the **Commitment Gap**: while the Global South talks about adaptation more than the Global North (higher keyword frequency), these commitments often lack the legal force of Northern mitigation laws. This reflects structural inequalities. Vulnerable countries cannot legally bind themselves to adaptation measures without guaranteed external finance.

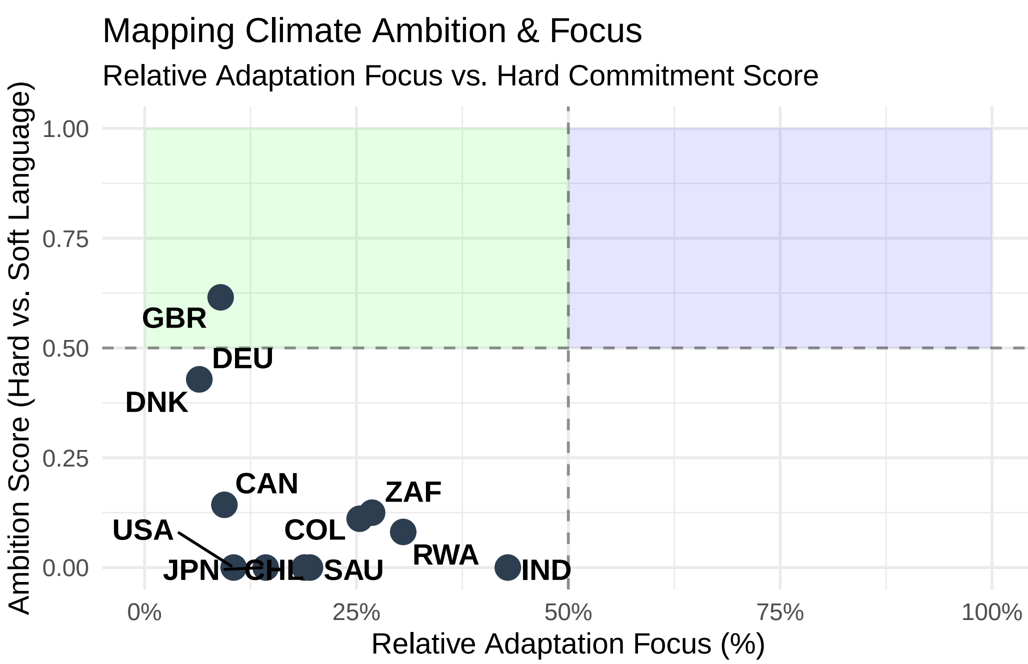

Figure 4: Climate Ambition Matrix: Lexical Hardness vs. Adaptation Focus

6 Limitations WIP

6.1 Limitation 1: The Persistent Invisibility of Adaptation

Challenge: Unlike mitigation's quantifiable carbon metrics, adaptation lacks systematic scrutiny. Measurement is a governance issue; wealthy nations have historically prioritized mitigation data (e.g., carbon accounting) to serve market interests. Reflections: The absence of standardized adaptation metrics in databases like the CPDB makes resilience "invisible" in policy debates and finance allocation. Future Work: Standardizing indicators, such as the percentage of the population covered by early warning systems, is essential for a balanced global audit.

6.2 Limitation 2: Missing Global South Data

Challenge: Global databases exhibit a geographical bias, offering higher resolution for the Global North while underrepresenting the Global South. CPDB primarily captures formal national legislation, often missing the informal, subnational, or community-based adaptation measures critical to resource-constrained settings. Reflections: My "Adaptation Mix" metric likely underestimates the true scale of resilience-building in the Global South due to this "streetlight effect" illuminating only what is easy to measure. Future Work: Future iterations should supplement global repositories with local government records and NGO reports to capture the full regulatory response.

7 Limitation 3: The Implementation-Reporting Gap

Challenge: Policy adoption does not equate to policy effectiveness. The Grantham Research Institute notes a severe lack of progress reporting; even where legal mandates exist, nearly half of the countries fail to publish regular assessments. Reflections: This "Commitment Gap" indicates weak enforcement and accountability. A country may lead in "Theory" (high policy volume) but fail in "Practice" due to resource constraints or political instability. Future Work: To assess true impact, policy data must be linked to outcome metrics like reduced disaster losses or improved food security.

8 Conclusion WIP

The three key findings—the Paris Inflection, the Survival Ratio, and the Commitment Gap—provide empirical evidence for climate justice debates: the Global South prioritizes adaptation, but lacks the resources to codify commitments into binding law.

The Grantham Research Institute's research confirms these patterns: "75% of finance-related laws and policies in the sample were introduced in Global South countries," "all 26 development plans that incorporate adaptation at a significant level originate from 13 countries in the Global South," and "between 2020 and 2024, more than 50% of the reviewed NDCs with a high emphasis on adaptation were submitted by Global South countries" [1, p.14-15].

9 References WIP

[1] Grantham Research Institute on Climate Change and the Environment, "Climate change adaptation laws and policies: Global trends and insights 2025," London School of Economics and Political Science, 2025.

10 LLM Declaration WIP

Word Count:

`Total Word Count: 2847`